

FIT3080

Assignment 2 – Knowledge representation and reasoning with Bayesian Networks [48 marks]

In this assignment, we explore the kind of AI which uses formal models to describe and make inferences about real-world systems, focusing on Bayesian networks.

The marks you earn in this assignment (out of a possible 48), including the associated Q&A component, will constitute 10% of your final grade for this unit. Please read this specification carefully, including the additional remarks and instructions on the final page.

Due Date: 11:55pm Monday, September 28, 2026 (Melbourne, Australia time)

Part 1: Distinguishing bacterial from viral pneumonia with Bayesian networks

Suppose that we want to build a (simple) Bayesian network model for determining if someone who presents to hospital with signs of pneumonia has bacterial or viral pneumonia, or no pneumonia.

Question 1 – Initial model structure (8 marks)

Design and build in Netica a causal 8-node BN structure (including nodes, arcs and possible states) that includes the following:

- A node to represent the patient's type of pneumonia, i.e., whether it is bacterial, viral or whether they have no pneumonia.
- A node to represent the patient's age in years, which influences what type of pneumonia a patient is susceptible to. Key age ranges are: under 15, 15-44, 45-64, over 64.
- A node to represent that the patient does or does not have a condition causing weakened immunity (immunocompromised) that makes a person more susceptible to bacterial infections.
- Two nodes to represent symptoms and signs of bacterial pneumonia: shortness of breath and the clinical blood marker C-reactive protein (CRP), which can be high (between 40 and 120), moderate (above 5) or low.
- A node to represent the result of an X-ray test, positive or negative, which provides an indication of pneumonia.
- A node to represent the results of a blood culture test, which can help identify if there is causative bacteria in the blood. The results of this test can be summarised as either positive (there was bacterial growth), negative (no growth) or inconclusive (potentially contaminated).
- A node to represent whether or not the patient is up to date with their pneumococcal vaccines, which protect against the main type of bacterial pneumonia.

For this question, you should only include arcs based on the information provided above.

Name your Netica BN file for this question: *FamilyName-StudentId-2ndSem2026FIT3080_A2_Q1.dne*

Include a screenshot of your BN in your written answer in the pdf file.

Question 2 – Parameterisation (10 marks)

You have access to multiple sources of information to parameterise your BN, including:

- ME: Information on risk factors from a medical expert
- LIT: Information on the reliability of test results from the literature
- DATA: Access to 3 different data sets covering signs and symptoms of pneumonia, test results and the influence of background factors

[ME] You obtained the following information from the ME on risk factors:

- The ME estimates 90% of all under 15s are up to date with their pneumococcal vaccination. Among those over 65, approximately 40% are up to date. In other age groups, due to lower risks, the proportion of up-to-date people is around 10%. Among immunocompromised individuals over 15, vaccination rates are 2x as high for over 65s, and 4x as high for other adult age groups.
- *If someone is up to date with their vaccinations (V) and not immunocompromised (not I),* their risk of bacterial pneumonia when presenting to hospital is 20% for patients under 15 years, 5% for adults, and 25% for those over 65. By contrast, if they are:
 - *V and I:* the risk is 50% higher.
 - *not V and not I:* the risk increases 2 fold.
 - *not V and I:* the risk increases 3 fold.
- 25% of under 65s presenting to hospital with an initial suspected pneumonia turn out not to have had pneumonia. This reduces to 15% for those over 65.
- 1% of those presenting to hospital with pneumonia are immunocompromised. For those over 65, this rises to 7%.

The ME also notes that immunocompromise can affect how CRP responds to pneumonia, and that age can moderate the effect of pneumonia on both CRP and shortness of breath. They provide no estimates, but you can obtain these from the DATA.

[LIT] You obtained the following information from the literature:

- A small study examined the ability to diagnose bacterial pneumonia from X-rays. The study showed that bacterial pneumonia is quite likely to lead to a positive test result on an X-ray, with an estimated 80% positive rate, compared to a 30% positive rate for viral pneumonia. The study did not investigate how often these findings appear for people with no pneumonia, but the authors noted that in their experience, this was quite rare, perhaps 2% of the time.
- A meta-analysis into blood culture testing showed that bacterial pneumonias give rise to a positive blood culture 85% of the time, but an inconclusive result 10% of the time. Both viral pneumonias and non-pneumonias lead to a positive blood culture around 15% of the time and an inconclusive result around 5% of the time.

[DATA] You have access to the following datasets for parameterisation:

- Data extracts that are relevant to your BN have been provided in the TRAIN zip file, available on Moodle. (Do not use the TEST data set to train your model.)

Revise and then parameterise your BN from Question 1 based on the information provided above, and provide a variable dictionary to document your BN. You will need to:

- i. Read through this question and make all the required structure changes

- ii. Parameterise your BN using the information from the ME and LIT
- iii. In preparation for training on data, weight the CPTs from the ME and LIT:
 - a. For ME CPTs, assign them the equivalent of 20 samples worth of information
 - b. For LIT CPTs, assign them the equivalent of 60 samples worth of information
- iv. Examine the TRAIN data .csv files (e.g., in spreadsheet software), and make your BN match the data (or vice versa) where needed. For continuous data, you can use continuous nodes and assign intervals so that Netica knows how to discretise the values from the data.
- v. Train the relevant parts of the BN on the 3 different data sets. Make sure to only select nodes that are in the data file when performing the training.
- vi. Create a variable dictionary for your parameterised BN (as a table) that includes:
 - a. The ID and/or label of the variable
 - b. A brief description of the variable
 - c. The states of the variable
 - d. Any parents of the node in the BN
 - e. Any information source that you used to parameterise the node (covering the ME, LIT or DATA)

Name your parameterised Netica BN file for this question:

FamilyName-StudentId-2ndSem2026FIT3080_A2_Q2.dne

Include the table and a screenshot of your BN in your written answer in the pdf file.

Question 3 – Reasoning (4 marks)

Use your Pneumonia BN (saved in Question 2) to answer the following questions (include screenshots from Netica for each scenario with your written answers):

- i. Suppose a 12 year old boy and a 72 year old woman present to hospital with suspected pneumonia at about the same time. Before you find out any of the other patient details, symptoms or test results, who is more likely to have bacterial pneumonia and by how much? What is the chance the boy has any type of pneumonia?
- ii. Suppose both present with shortness of breath. The boy also has a CRP of 65, while we are not yet aware of what the woman's CRP levels are.
 - a. What is the woman's predicted CRP level (as a mean and standard deviation)?
 - b. What is the probability of bacterial pneumonia for each patient?
 - c. What is the probability that the boy is immunocompromised before and after observing CRP? What explains the direction of change?
- iii. Suppose they both have an X-ray. The woman's X-ray is positive, while the boy's is negative. Who is more likely to have bacterial pneumonia? What is the probability that each patient has viral pneumonia?

The answer to this part should appear in the pdf file.

Question 4 – Sensitivity and performance (5 marks)

Evaluate your BN using the sensitivity and performance testing tools that Netica provides.

(Note: You will not be marked on the performance of your BN, only on whether you perform the evaluation correctly.)

- i. Generate a sensitivity table for the pneumonia node with no evidence in the network.
 - a. What is the entropy of the pneumonia node when there is no evidence in the network?
 - b. Which variable would provide the most information about pneumonia on its own?
 - i. What is the mutual information of this variable with pneumonia?
 - ii. How much do we expect the entropy of pneumonia to reduce if we observed this variable?
 - c. Which variable provides the *least* information about pneumonia? What is its mutual information with pneumonia? Why is the mutual information lowest for this variable?
- ii. You observe a negative X-ray result for a patient who is immunocompromised.
 - a. Which variables now provide more information than before? Which variables provide less? Why might some variables now have less mutual information with pneumonia?
- iii. Use the 'TEST_DATA' data set from Moodle to test the performance of your BN in predicting pneumonia.
 - a. What is the error rate of your model on this TEST data?
 - b. What is the confusion matrix? Where does it perform worst?
 - c. What is the log score?
- iv. Create 1) a version of your BN with the tests removed (i.e., remove X-ray result and blood culture result, while leaving everything else the same) and 2) a one-node BN with just the pneumonia node from your model. For the one-node BN, make sure the probabilities match the marginal probabilities of pneumonia in your full BN. Run the tests from (iii) on both models (1) and (2).
 - a. How do the error rates, confusion matrices and log scores for both (1) and (2) compare to the results from (iii)?
 - b. What does this tell you about the performance of your full model?

The answer to this part (including the sensitivity table and requested performance test results) should appear in the pdf file.

Question 5 – Decision-making model structure (5 marks)

Next we want to extend your BN into a Bayesian decision network (BDN) to help with decisions as to whether **to perform an X-ray or not** and whether **to treat with antibiotics or not** (which is able to treat the bacterial pneumonia, but not viral pneumonia) to reduce the risks associated with a severe bacterial pneumonia. You want the BDN to be able to represent the cost (and harm) of doing the X-ray and the cost (and harm) of the antibiotic treatment, as well as the benefit to the patient with a bacterial pneumonia of treating with antibiotics.

- i. Extend your BN with 2 decision nodes, and 3 utility nodes, connected to the chance nodes in your existing network. You will need to add a fourth state (“no result”) to the X-ray test node and adapt the CPTs for it accordingly. Include at least 1 information link, which points into the decision node for treating with antibiotics.
- ii. Create utility tables for your utility nodes so that the best choice for someone under 15 with low CRP (i.e., < 5) to have an X-ray (i.e., this choice should lead to the highest expected utility) is determined as follows.
 - If their X-ray is negative, their highest utility should then correspond to not being given antibiotic treatment.
 - If their X-ray is positive, their highest utility should correspond to being given antibiotic treatment.

Utilities should reflect the value of all the harms and benefits (not just costs) and also that untreated bacterial pneumonia causes more harm than viral pneumonia.

Include the utility tables (screenshots) in your written answer in the pdf file, and explain why you have chosen those values.

Name your extended parameterised Netica BDN file for this question:

FamilyName-StudentId-2ndSem2026FIT3080_A2_Q5.dne

Question 6 – Decision-making (4 marks)

- i. What are the expected utilities computed by your BDN for the DoXray decision (i.e., doing the X-ray test, or not doing it) for someone under 15 with low CRP before they have had an X-ray? What are the expected utilities for the Antibiotics decision (i.e., treat or don't treat with antibiotics), given each possible choice of the DoXray decision?
- ii. If you learn that they have shortness of breath, how does that change the expected utilities for DoXray? What if their CRP was also moderate? Why might it not be worth doing an X-ray in this case, and what Antibiotics decision would you make without the X-ray result?

The answer to this question should appear in the pdf file.

Include in the pdf file a screenshot of your BDN in Netica with evidence entered for these scenarios and showing the expected utility values.

Part 2: Question and Answer – In-class Reflection [12 marks]

The Q&A component contributes 12 marks of the total assignment marks and will be completed during your scheduled applied class. This component is designed to assess your understanding of your own solution.

During the session, you will answer a set of written questions about your submitted assignment. The questions are intended to help your tutor understand:

- your overall approach to solving the problem;
- the rationale behind your design decisions; and
- how you interpret and evaluate your results.

The questions are **not designed to trick you**. If you have completed the assignment yourself and understand the design decisions you made, you should be well prepared to answer them.

Format

- The Q&A will be conducted in your scheduled applied class.
- Responses must be completed by hand on paper.
- Computers, electronic devices, and online resources are not permitted during the Q&A unless previously authorised.
- The questions will focus on your submitted assignment and your understanding of BNs.

Assessment Criteria

Your responses will be assessed holistically based on your ability to:

- clearly explain and justify your design decisions; and
- demonstrate an understanding of BNs and their evaluation.

Additional details are provided in Table 1.

Table 1: Marking rubric for Q&A component.

Performance	Design	Evaluation
Excellent	Provides clear, well-justified explanations for design choices and demonstrates a strong understanding of BNs.	Interprets results accurately, and provides meaningful insights.
Good	Justifies most design decisions and demonstrates a sound understanding of BNs.	Provides a reasonable interpretation of the results.
Satisfactory	Describes the design choices with limited justification.	Summarises the results with limited analysis.
Unsatisfactory	Provides little or no explanation of design decisions.	Shows little understanding of the obtained results.

SUBMISSION INFORMATION

Due Date: 11:55pm Monday, September 28, 2026 (Melbourne, Australia time)

Method of submission

Your submission should consist the following files:

- A text-based .pdf file, with screenshots as necessary, named: *FamilyName-StudentId-2ndSem2026FIT3080_A2.pdf*
- A .dne file named *FamilyName-StudentId-2ndSem2026FIT3080_A2_Q1.dne*
- A .dne file named *FamilyName-StudentId-2ndSem2026FIT3080_A2_Q2.dne*
- A .dne file named *FamilyName-StudentId-2ndSem2026FIT3080_A2_Q5.dne*

Questions should be answered in both the .pdf and the .dne files. The requested screenshots should be included in the .pdf file.

All the files must be uploaded on the FIT3080 Moodle site by the due date and time.

Sources of information

You can use background material to solve this assignment, e.g., videos or lectures on Bayesian networks and any other information on building, parameterising and testing Bayesian networks.

Plagiarism

INTENTIONAL AND UNINTENTIONAL PLAGIARISM IS NOT ACCEPTABLE,
AND WILL HAVE CONSEQUENCES IF YOU ARE DEEMED TO HAVE CHEATED.